

Climate 5000 Series - Gen 4

60K BTU Single Zone Regular Light Commercial System

Ducted IDU

Engineering
 Submittal
 Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

60K Btu Single Zone Light Commercial Regular System - Ducted IDU

Indoor part number		8-733-962-715
Indoor model number		BMS500-AAU060-1AHDXD
Outdoor part number		8-733-962-694
Outdoor model number		BMS500-AAS060-1CSXLD
Efficiency Data		AHRI 210/240 2023
SEER2		17.0
EER2		10.0
HSPF2 Region IV		9.7
HSPF2 Region V		7.6
COP(47F)		3.70
COP (5F)		2.20
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	14600~63000
	Rated Capacity	58000
	Current	25.6
	Input Power	5800
Heating	Capacity Range	11800~67000
	Rated Capacity (47F)	60000
	Rated Capacity (5F)	40000
	Current	21.0
	Input Power	4750
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		5.0 / 42.0
Max. fuse indoor/outdoor (A)		15.0 / 45.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 1400
Input (W)		3700
Rated current (RLA/A)		7.0
Piping		
Refrigerant precharge (oz/kg)		144.62 / 4.10
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		3/8" / Ø9.52
Gas side (in/mm)		3/4" / Ø19.0
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.32 / 30
Max. refrigerant pipe length (ft/m)		246.0 / 75
Max. difference in height (ft/m)		98.4 / 30

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Operation Range

ODU	Cooling (°F / °C)	-13 ~ 122 / -25 ~ 50
	Heating (°F / °C)	-13 ~ 75 / -25 ~ 24
IDU	Cooling (°F / °C)	60 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

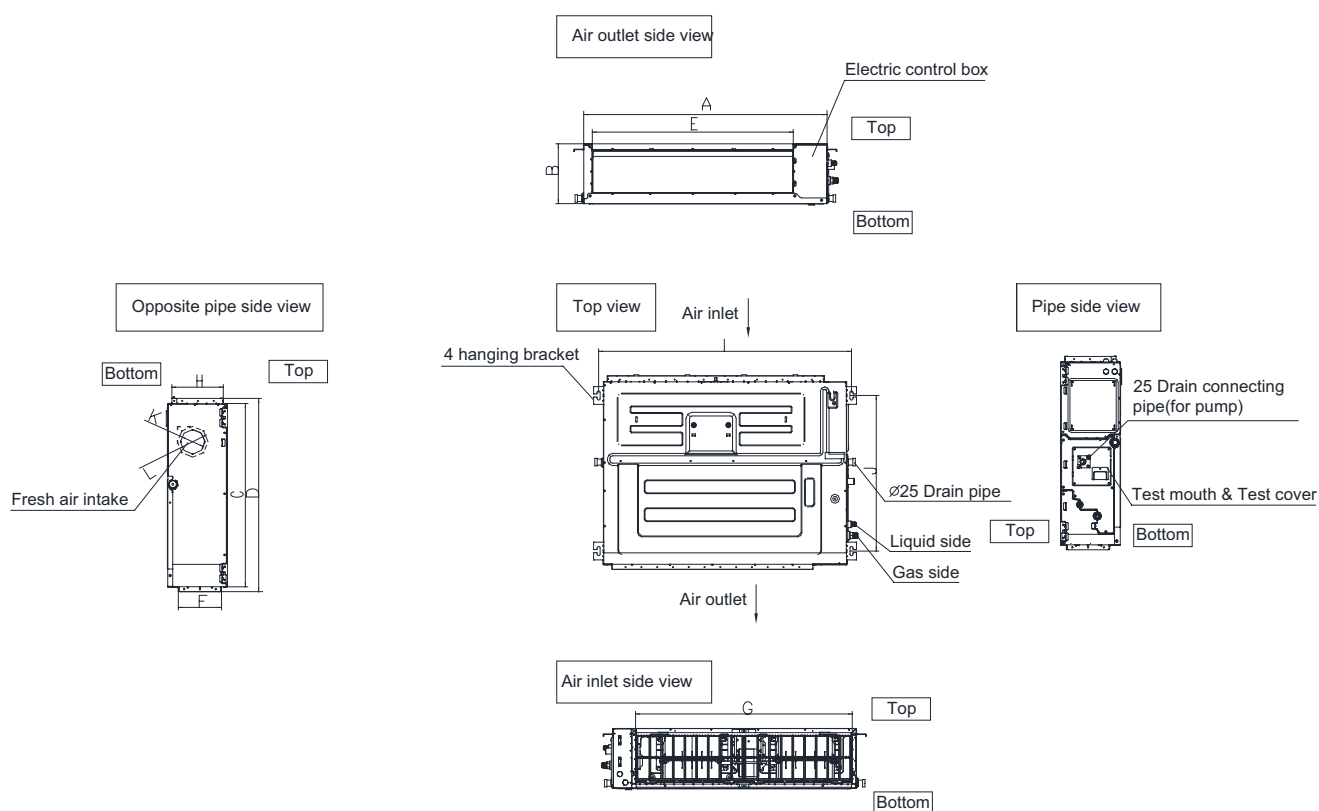
System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		11.07
IDU	Fan motor model	ZKFN-560-8-4
	Fan motor qty (ea)	1
	Fan motor input (W)	665.0
	Fan speed, H / M / L (RPM)	850 / 780 / 700
	Air flow, Turbo / H / M / L / Si (CFM)	2001.2 / 1706.9 / 1412.6 / 1118.3 / N/A
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 52 / 48 / 43 / 42
	Net weight (lbs / kg)	161.38 / 73.2
	Gross weight (lbs / kg)	178.13 / 80.8
ODU	ESP range (Pa)	0-200
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-85-8-22-5
	Fan motor qty (ea)	2
	Fan motor input (W)	126
	Fan motor speed, H / / L (RPM)	850 / 750 / 650
	Air flow, Max (CFM)	4237.9
	Noise level, Max (dB(A))	64.0
Accessories	BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
	BMS-DRY-XXD (8733969263)	24V Interface

Bosch Thermotechnology Corp.
 Londonderry, NH • Watertown, MA • Ft. Lauderdale, FL

Tel: 1-800-283-3787 www.bosch-homecomfort.us

Indoor Unit Dimensions and Connections

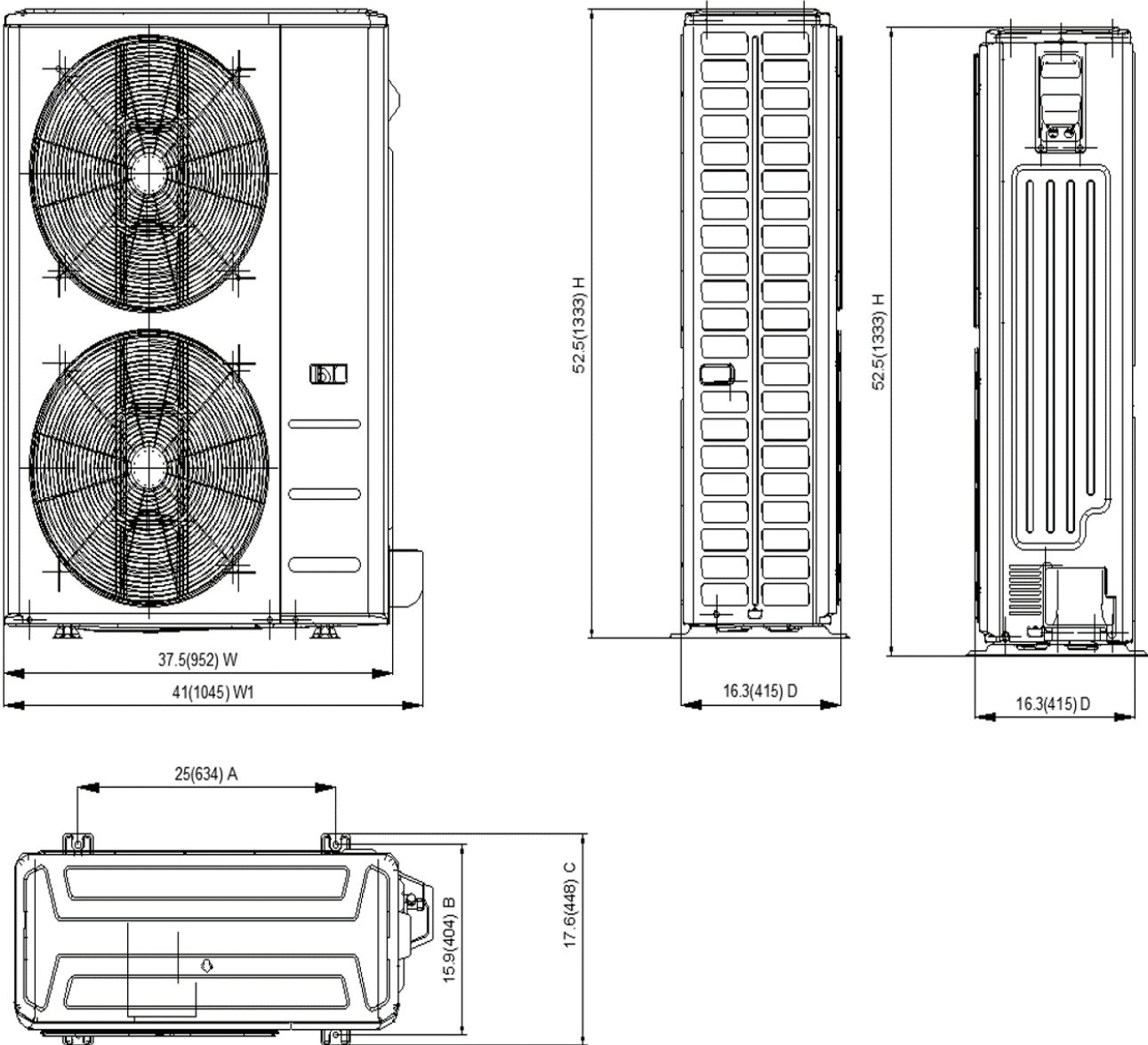


Model	Outline Dimension (in)				Air Outlet Opening Size (in)		Air Return Opening Size (in)		Size of Mounted Lug (in)		Fresh Air Intake Opening Size (in)	
	A	B	C	D	E	F	G	H	I	J	K	L
BMS500-AAU060-1AHDXD	55.1	14.9	31.5	33.3	48.1	12.6	50.1	13.0	56.7	26.3	4.9	6.3

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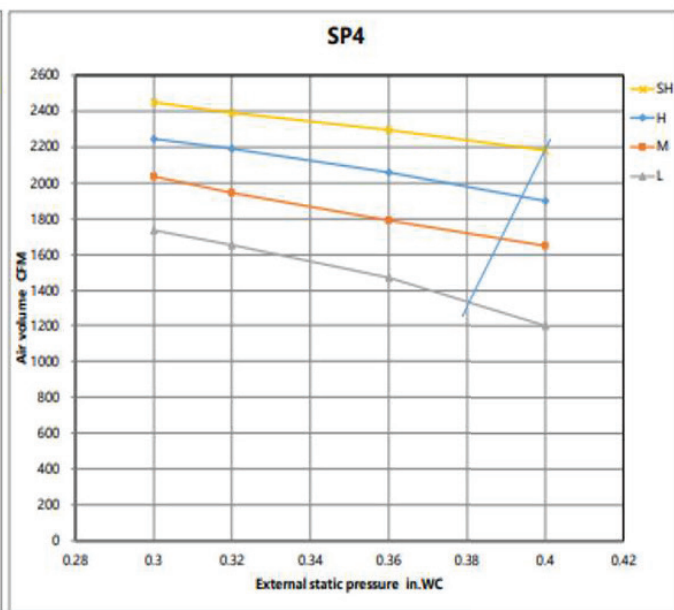
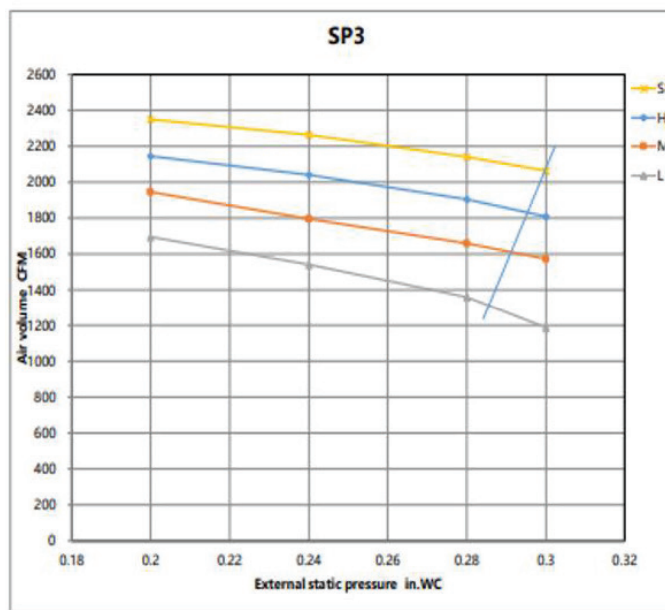
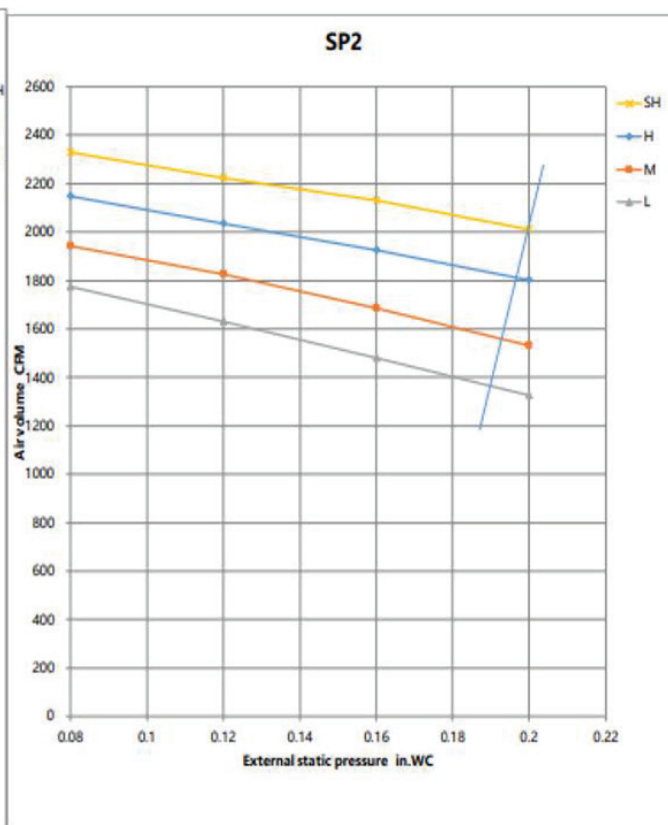
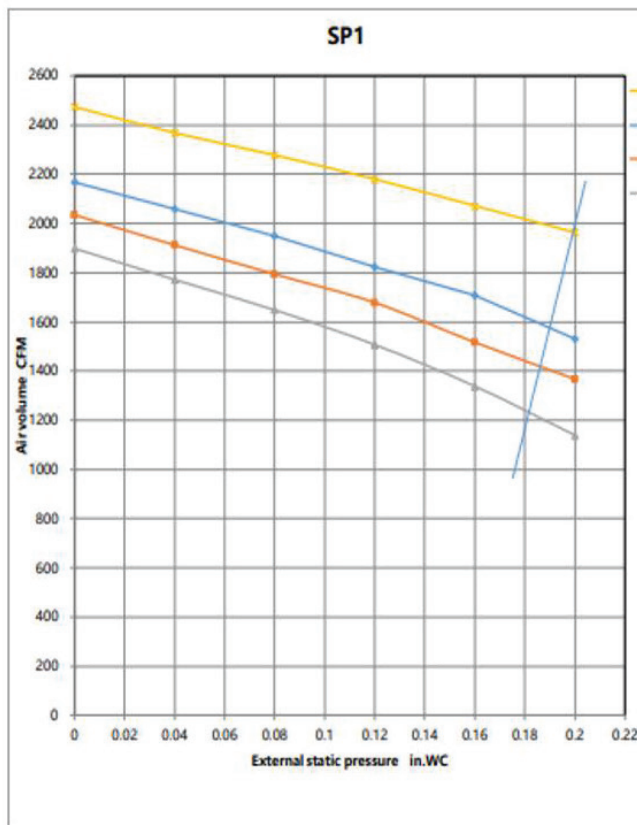


Outdoor Unit Dimensions and Connections

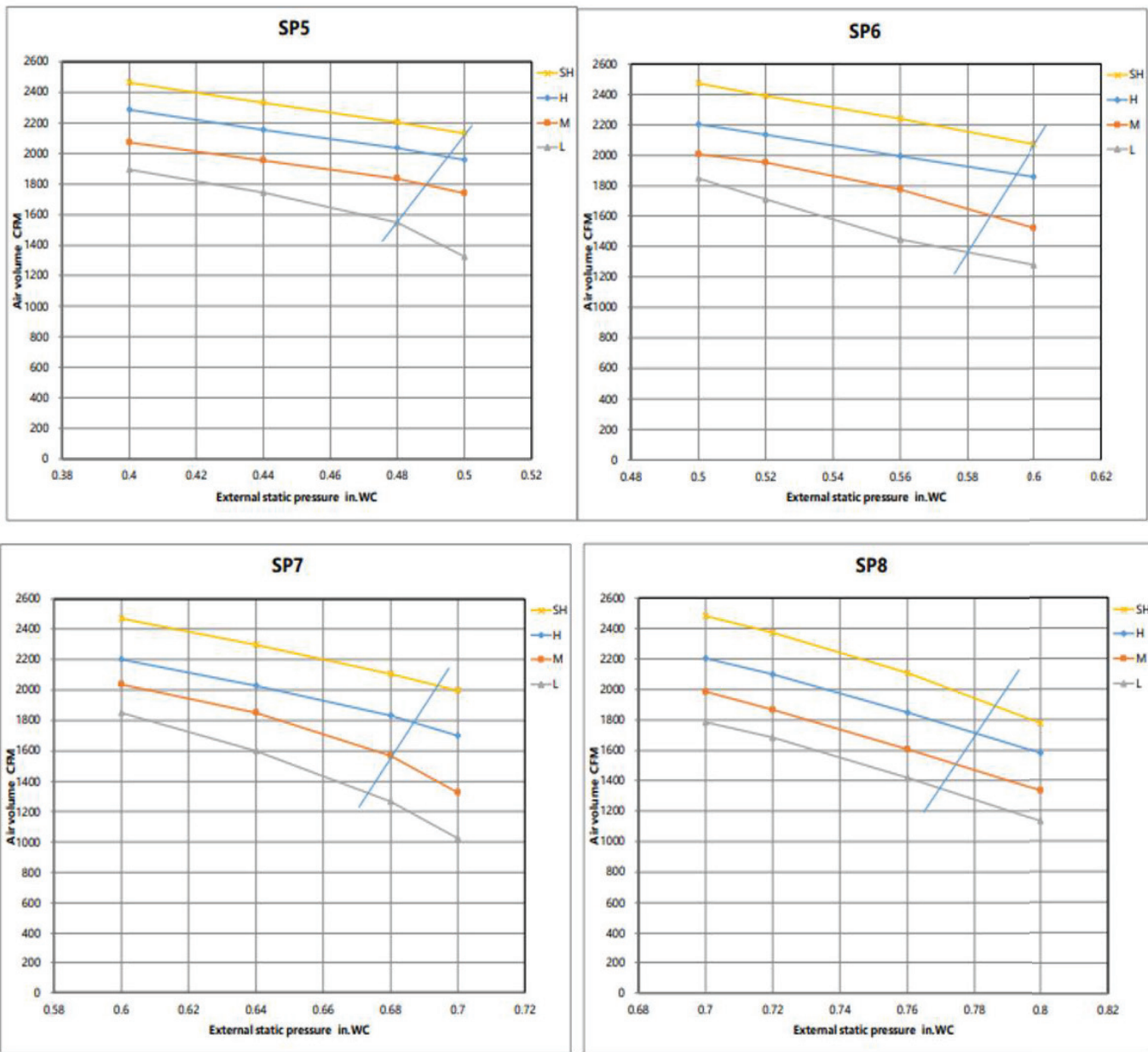


Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)
BMS500-AAS060-1CSXLD	37.48	16.34	52.48	41.0	37.48	25.0	15.9	17.6

Air Volume

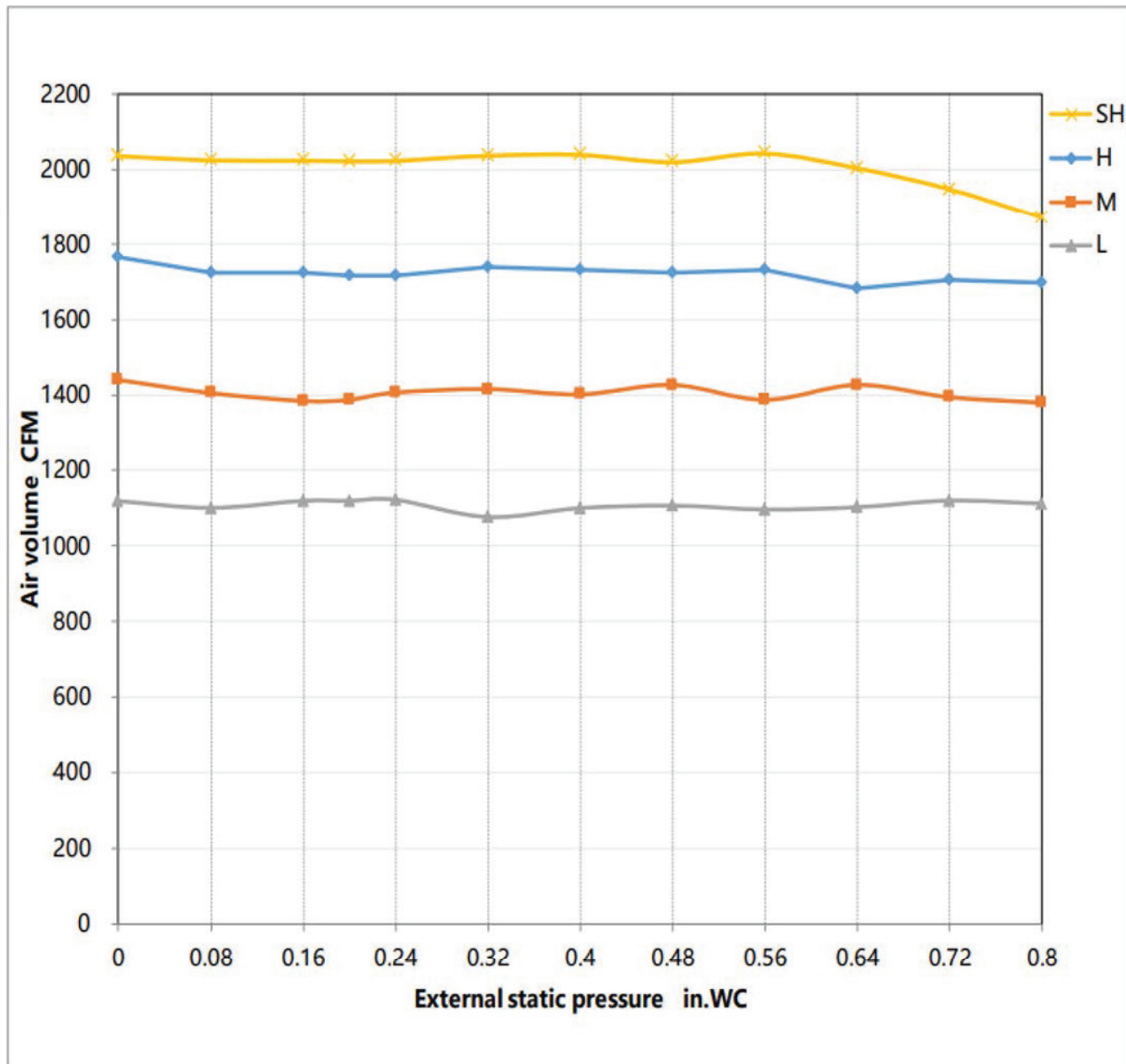


Air Volume



Air Volume

Constant air volume



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Heating Performance - Indoor Dry Bulb Temperature at 2000 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	24847	24350	24229	24132	23915	4.32	4.35	4.36	4.37	4.39
-4°F(-20.0°C)	32682	32028	31868	31741	31455	5.02	5.06	5.07	5.08	5.10
0°F(-17.8°C)	36388	35660	35482	35340	35022	5.12	5.16	5.17	5.18	5.20
5°F(-15.0°C)	41020	40200	39999	39839	39480	5.24	5.28	5.29	5.30	5.33
14°F(-10.0°C)	41889	41051	40846	40683	40316	4.99	5.03	5.04	5.05	5.07
17°F(-8.3°C)	42178	41335	41128	40964	40595	4.91	4.95	4.96	4.97	4.99
22°F(-5.6°C)	44990	44091	43870	43695	43301	5.06	5.11	5.12	5.13	5.15
27°F(-2.8°C)	47802	46846	46612	46426	46008	5.22	5.26	5.27	5.29	5.31
32°F(0.0°C)	50614	49602	49354	49157	48714	5.38	5.42	5.43	5.44	5.47
35°F(1.7°C)	52301	51255	50999	50795	50338	5.47	5.52	5.53	5.54	5.56
37°F(2.8°C)	54972	53872	53603	53389	52908	5.41	5.46	5.47	5.48	5.50
42°F(5.6°C)	61648	60415	60113	59872	59334	5.27	5.31	5.32	5.33	5.35
44.6°F(7.0°C)	64318	63032	62717	62466	61904	5.21	5.25	5.26	5.27	5.29
47°F(8.3°C)	68324	66957	66623	66356	65759	5.12	5.16	5.17	5.18	5.20
52°F(11.1°C)	66548	65217	64891	64632	64050	4.61	4.64	4.65	4.66	4.68
57°F(13.9°C)	64773	63477	63160	62907	62341	4.09	4.13	4.13	4.14	4.16
59°F(15.0°C)	64062	62781	62467	62217	61657	3.89	3.92	3.93	3.93	3.95
62°F(16.7°C)	62997	61737	61428	61183	60632	3.58	3.61	3.61	3.62	3.64
64.4°F(18.0°C)	62145	60902	60597	60355	59812	3.33	3.36	3.36	3.37	3.38
69.8°F(21.0°C)	55241	54136	53865	53650	53167	2.77	2.80	2.80	2.81	2.82
75.2°F(24.0°C)	48333	47367	47130	46941	46519	2.22	2.24	2.24	2.25	2.25

Bolded values were tested. All other values are simulated and for reference only.